

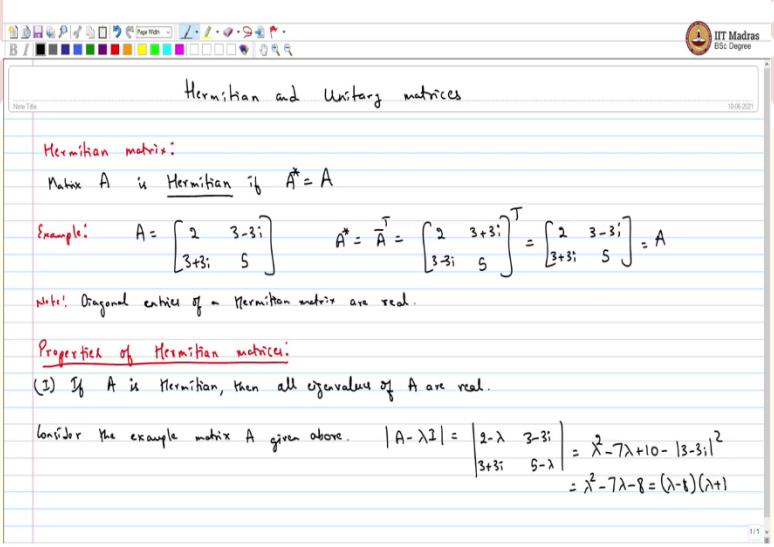
IIT Madras

ONLINE DEGREE

Machine Learning Foundation
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Hermitian Matrices

Welcome to the second lecture in week 5 of Machine Learning Foundations. In this week, as I have said before, we will be focusing on Hermitian matrices. So, we will continue the discussion of Hermitian matrix, matrices. This class of matrices was defined in the previous lecture, and in this lecture we will state and prove two important properties concerning the eigen values and eigen vectors of a Hermitian matrix.

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Welcome, to another lecture of linear algebra, module of Machine Learning Foundations. Today we will be looking at two important class of matrices, Hermitian and Unitary matrices. So, let us *t with the first one. So, what is a Hermitian matrix? So, matrix A is Hermitian if its conjugate transposes itself. So, let us look at a quick example, what this means. Look at the matrix, suppose A is this, 2 5 3 - 3i and 3 + 3i. So, what is A *? A * is the conjugate transpose = 2 3 + 3i 3 - 3i and 5. This is the conjugate. And then you have to take transpose. And that turns out to be just A.

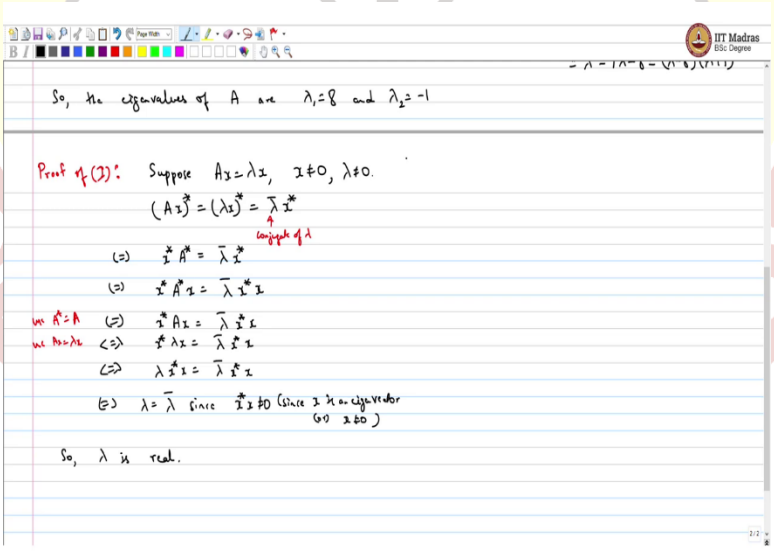
So, this is an example of a conjugate transpose. You could do it in the other way also. First you could take transpose and then take the conjugate. The operations are interchangeable, and you will get the same result. And Hermitian matrices are a very important class of matrices because they constitute a set of orthogonally diagonalizable matrices.

We will look at that very soon, but let us understand Hermitian matrices a little more. I mean you can observe that, note, diagonal entries of a Hermitian matrix are forced to be real valued. So, you cannot have complex values along the diagonal, and then hope to get a Hermitian matrix. Now, there are some important properties of Hermitian matrices which we will look at. Let me state them as a theorem, and prove them one by one. Let us call this as properties, properties of Hermitian matrices.

The first important property is, if a matrix A is Hermitian then all eigen values of A are real, real value, they cannot be complex. Let us look at the same example to see what the eigen values are. I mean, let us look at an example first, before proving, this consider the example matrix A given above.

For this matrix, if you do, if you calculate the eigen values, I would suggest you pause the video now, lecture video, and calculate the eigen values on your own. I will just do this in here, 2 - λ 3 - 3i 3 + 3i 5 - λ. The determinant is actually, it turns out to be λ square - 7 λ + 10 - mod 3 - 3i square, which is the same as λ square - 7, 7 λ - 8, this is the same as λ - 8 into λ + 1.

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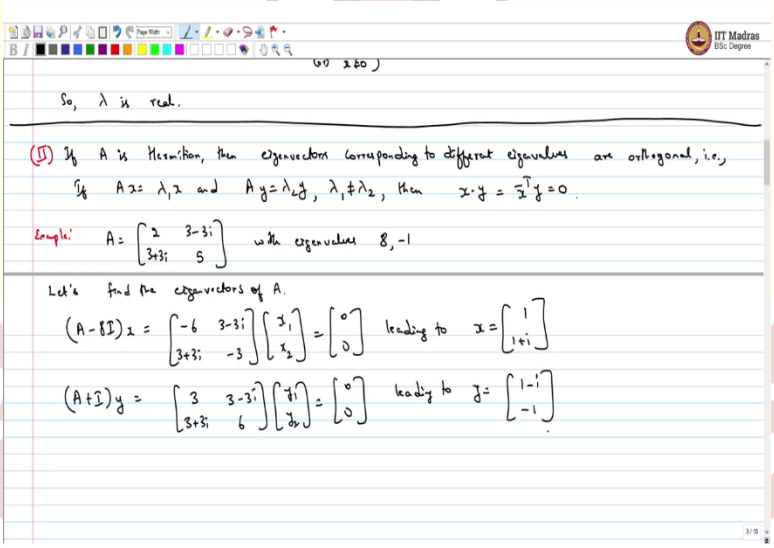
So, so the eigen values of A are λ 1 = 8 and λ 2 = - 1. Now, it did not turn out by accident that these are real. I mean in general for any, in general for any Hermitian matrix, it is always true that eigen values are real. So, we will just prove this. Proof so 1. So, let us *t

with the usual eigen equation. Suppose A x = λ x. Obviously just to emphasize you cannot, x which is the eigen vector, has to be non-trivial, looking at a non-zero eigen value.

Now, to prove, to prove that λ can only be real, it is enough to show, that if I take the conjugate of λ, I recover λ. So, λ = λ bar. To do that let us *t by taking the conjugate of Ax *. This is the conjugate of λ x, this by the rules of conjugation is λ bar x *.

So, here, conjugate of λ. Now, so this, equivalently A x, * by rules of conjugation, is x * A * = λ bar x *. This is what we have. Now, let us multiply x on both sides, λ bar x * x. And now let us use the fact that A is Hermitian. So, I get x * Ax = λ bar x * x. But this is a special x as A x = λ x. So, what do I get? I get x * λ x = λ bar x * x. And this just means, implies λ × x * x = λ bar × x * x, and this finally =, equivalent to saying λ = λ bar since x * x is not 0, since x is an eigen vector, or x ≠ 0. So, λ is, that is the end of this claim. I hope this is clear.

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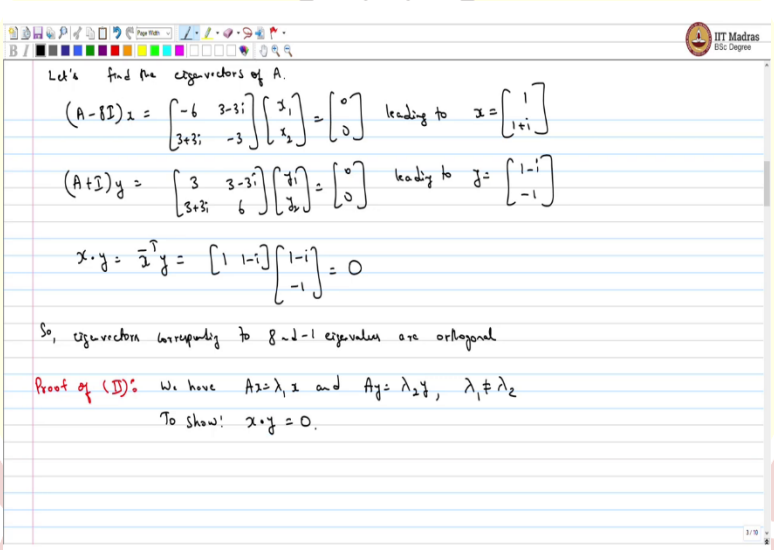


Now, we look at the second property, Property 2 of a Hermitian matrix, which says if A is Hermitian, then eigen vectors corresponding to different eigen values are orthogonal, that is, Ax = λ 1 x Ay = λ 2 y, λ ≠λ 2, then x . y, which is x bar transpose y is equal to 0. So, that is the . product in a complex vector space. As before, we could look at the example. So, let us look at the example before. Recall what we had as A is 2 3 - 3i 3 + 3i and 5, with eigen values, which we calculated as 8 , - 1.

Now, what are the eigen vectors? Again, let us find the eigen vectors. So, first thing is I want A - the first eigen value, so A 8 i - × any vector x, so this is 2 - 6 is -, no 2 - 8 is - 6, 3 - 3i 3 + 3i and - 3, this into let us say x 1 into x 2, I want this to be equal to 0. And you can work out the solution on your own leading to x = 1 , 1 + i. So, this is eigen vector 1.

Let us look at the second eigen vector. A + i into y, let us call this y. So, this would be 3, 3 - 3i, 3 + 3i and 6, this into say y1 y2 = 0 0, and this leading to y = 1 - i - 1. It is a simple enough system so I m just guessing the eigen vectors. You could go about doing this in a more methodical way, for larger matrices using Gaussian elimination, but given that, given that this is a simple 2 cross 2 matrix, just by eyeballing you will be able to guess the eigen vectors.

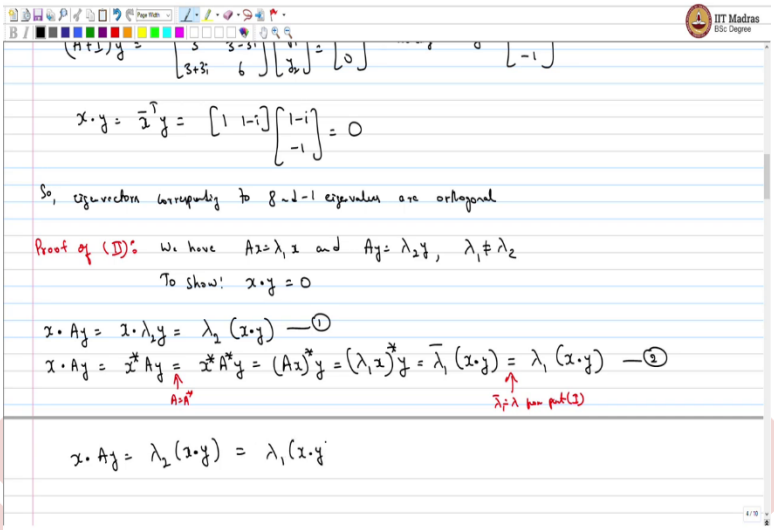
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Now, if you look at x . y, which is x bar transpose y, this is 1 ,, x bar, is 1 , 1 - i and then you transpose it, and take its . product with this, and this is clearly 0. So, in this example it is clear that, so eigen vectors corresponding to 8 and - 1 eigen values are orthogonal. And this is again not an accident. This holds true in general. Just illustrated using an example.

So, let us look at the proof of the second claim in the most general case. So, again, so we have, we have Ax = λ x, so λ 1, and Ay = λ 2y, and we are given that λ 1 ≠ λ 2. Now, what do we have to show? To show x . y is 0.

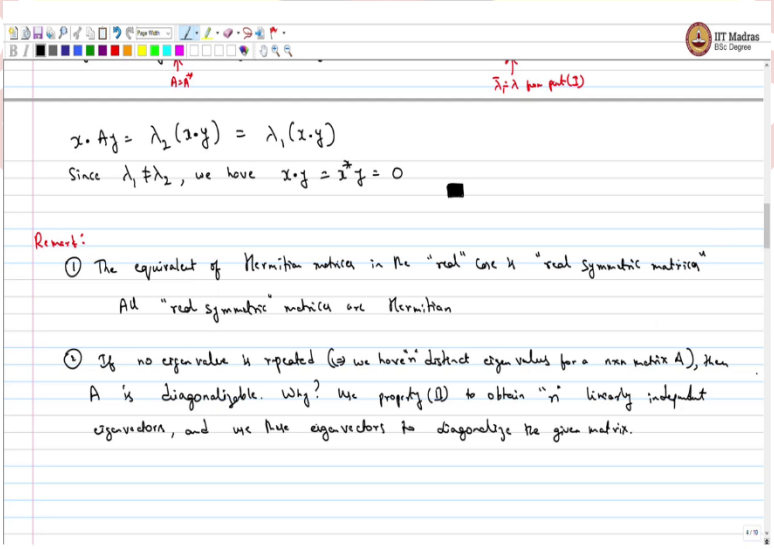
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And how do we show, do this through x . Ay. This is x ., simply because Ay = λ 2 y, and this is equal to λ 2 into x . y. Second way of doing the same thing is x . Ay is x * Ay, that is just the definition, x * is x bar transpose. But we also know that A = A *.

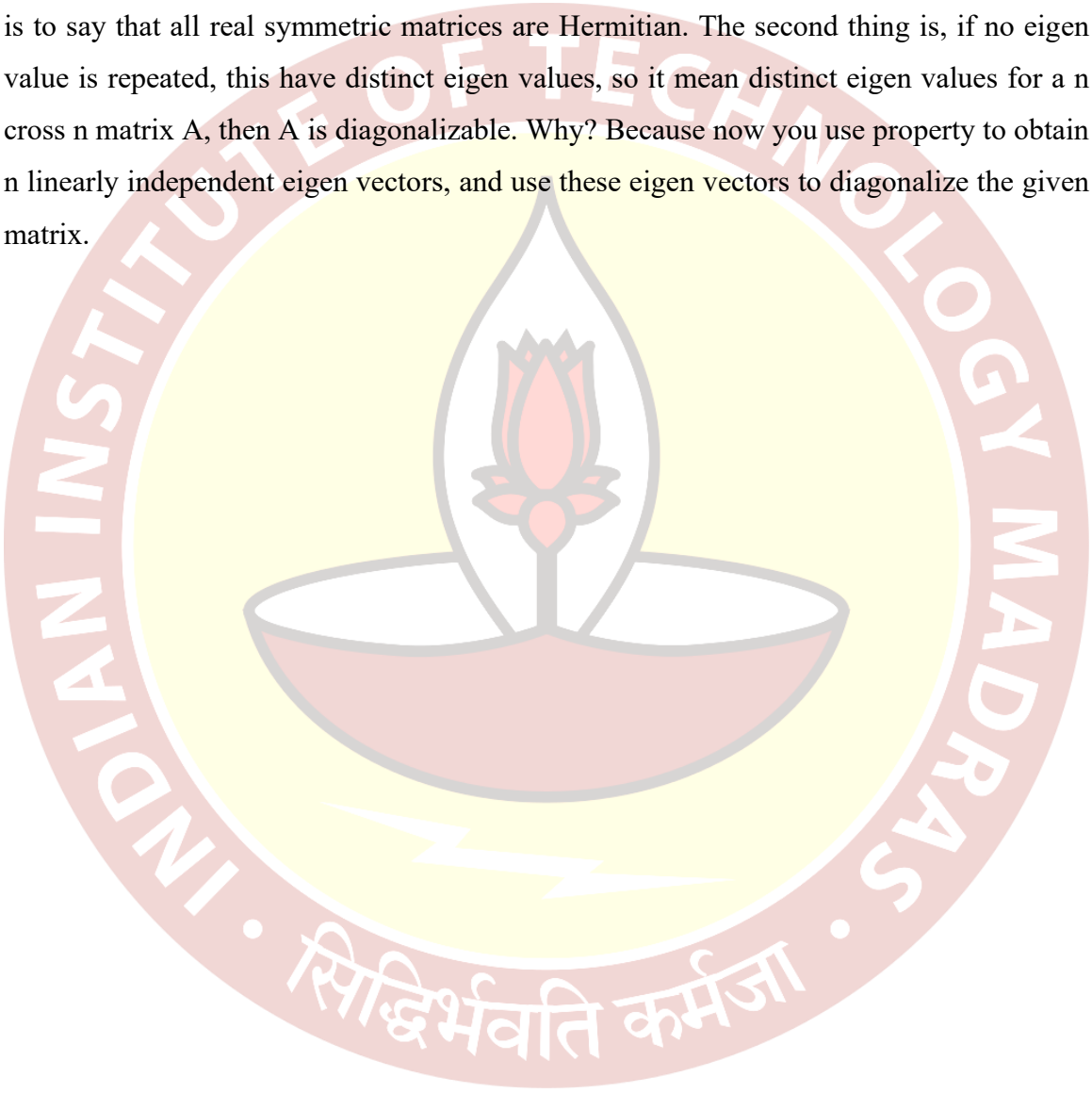
So, I can write this as x * A * y, which is the same as Ax * y. But Ax is λ 1 x * y. But λ 1 comes out so it will be λ 1 bar × x . y. Now, this is equal to λ 1 × x . y. Now, let me see, because here A = A *. And here we used λ 1 bar = λ from part 1, because it is a Hermitian matrix. So, the net effect, what we got is, let me call this equation 1, equation 2. So, x . Ay = λ 2 × x . y, and that is also equal to λ 1 × x . y.

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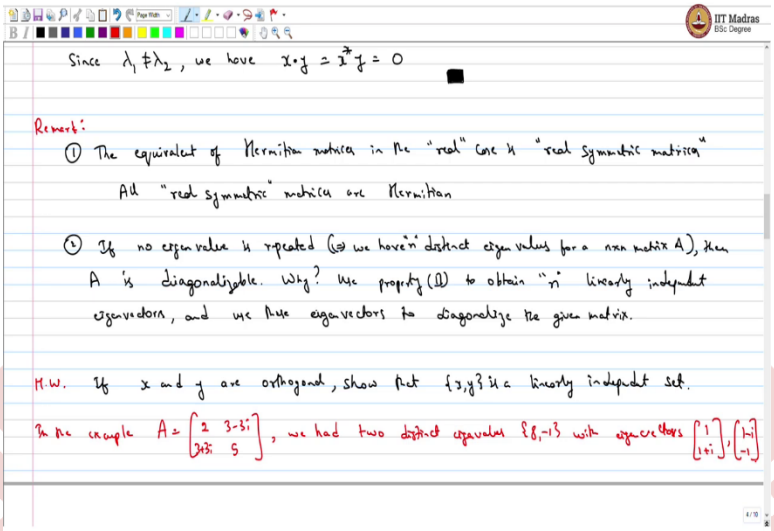


Since $\lambda 1 \neq \lambda 2$, we have, the only way I can get this equality is that I get x . y = x * y = 0. And this is the end of the proof. It is the end of this proof. Please go over this carefully, and if I go back to this example, you can observe, let us, let me make a remark, remark, the equivalent of Hermitian matrices here is real symmetric matrices.

There is no question of conjugation. You just take the transpose. So, trivially, I mean, that is to say that all real symmetric matrices are Hermitian. The second thing is, if no eigen value is repeated, this have distinct eigen values, so it mean distinct eigen values for a n cross n matrix A, then A is diagonalizable. Why? Because now you use property to obtain n linearly independent eigen vectors, and use these eigen vectors to diagonalize the given matrix.



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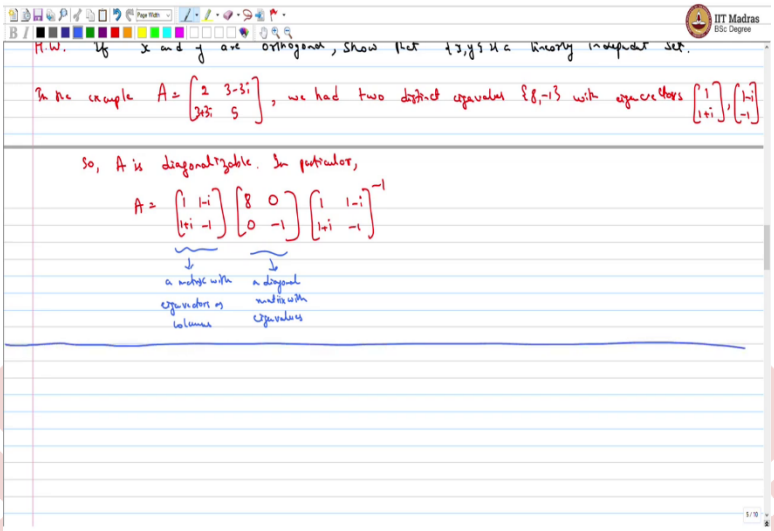


There are two things that, so the point is we just showed that if, if eigen values are different, then the corresponding eigen vectors are orthogonal, and orthogonal vectors are linearly independent. And this is something I must have shown before. Think about this. I am not sure entire, homework, if x and y are orthogonal, show that x , y is a linearly independent set.

It is a fairly straightforward thing to do. I mean, think about that. It is fairly straightforward. I can give you a overall hint. So, if you want x and y, a linear combination to be 0, you have to look at linear combinations of x and y that lead to 0, say alpha 1 x1 + alpha 2 y = 0, and then take a . product, take a . product with anything. So, you say x, to infer the other (cons) other coefficient is 0, say alpha 2 = zero, and do the reverse. And that is, that is a very strong hint. Please think about and solve this.

So, property 2 says that, I mean, if I have n distinct eigen values, from property 2 I can infer that I will have n linearly independent eigenvectors and I can stack these eigen vectors in a matrix and diagonalize it. That is really the claim that we are trying to make here. So, in the example, where A was 2 3 - 3i 3 + 3i and 5, and we had two distinct values 8 - 1 with corresponding eigen vectors 1 , 1 + i 1 - i , - 1.

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So, A is diagonalizable. In particular, way to diagonalize this is, you can write A as, first put a diagonal matrix with the eigen values, so the idea of diagonalization is you stack, so this, the idea is matrix with eigen vectors as columns and this is a, a diagonal matrix with eigen values along the column, along the diagonal, and this is the inverse of the first matrix. So, this is how you diagonalize it.

So, that brings us to the end of Hermitian matrices. We will revisit this in the context of diagonalization by proving that all Hermitian matrices are diagonalizable. A quick summary before I wrap up this lecture. In this lecture, we discussed two important properties of a Hermitian matrix, which namely, the first property states that all the eigen values of a Hermitian matrix are real.

And the second property is, if you have two different eigen values, distinct ones, then their corresponding eigen vectors are orthogonal. Now, this led to a very easy conclusion that if I have n distinct eigen values for a n cross n Hermitian matrix, or any matrix, then one can easily infer that it is diagonalizable because I get n orthogonal vectors, and if vectors are orthogonal then they are also linearly independent.

So, I get enough vectors to diagonalize the matrix. So, that is a easy case we can cover, but in general Hermitian matrices need not have distinct eigen values. Even in that case we will actually proceed to show in forthcoming lectures that they are diagonalizable even if some eigen values get repeated. That is the, the claim that we are trying to get to finally.